

ECO387C - Fall 2024

Discussion 14

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In these notes I present a fairly general cookbook recipe of how to solve a large class of dynamic general equilibrium models without uncertainty. I do so by using a version of the Solow model as an example. Broadly speaking the steps are the following:

1. Model description and equilibrium definition
2. Solution
 - (a) Solving every agents' optimization problems
 - (b) Imposing market clearing
 - (c) Exploring other properties of the model (here: in particular dynamics and the interpretation of depreciation)

At the very end I solve explicitly the nonlinear differential equation that arises from Solow-type models.

1 Model description and equilibrium definition

Consider a continuous time Solow model with investment-specific technological change. The model is characterized as follows

- The aggregate production function is

$$Y_t = X_t^\alpha L_t^{1-\alpha}, \quad \alpha \in (0, 1).$$

Note that the key difference between this and the basic Solow model is that X_t is not the market value of capital but the physical quantity of capital (machines).

- The capital accumulation equation is

$$\dot{X}_t = I_t^X - \delta X_t, \quad \delta \in (0, 1).$$

Here, of course, I_t^X denotes the physical quantity of capital per time unit that is added to the capital stock.

- The characterizing property of Solow-type models is the constant savings rate: we assume that households save a constant share of their income Y_t so that $S_t = sY_t$.
- It is assumed that there is technological progress in producing the capital good over time. In particular, we assume that at time t one unit of GDP can be transformed into q_t units of the capital good, or $I_t^X = q_t I_t$ and that q_t grows exponentially:

$$\dot{q}_t = \gamma q_t, \quad \gamma > 0.$$

- The population grows at rate $n > 0$, or

$$\dot{L}_t = nL_t$$

I will assume that there are three different types of agents in this economy: consumers, output-producing firms and capital goods-producing firms.

Definition 1 *A competitive equilibrium in this economy is a time path of allocations $\{Y_t, S_t, C_t, I_t, I_t^X\}_{t=0}^\infty$ and the capital stock $\{X_t\}_{t=0}^\infty$ and factor prices $\{w_t, R_t\}_{t=0}^\infty$ and a capital price $\{p_t^X\}_{t=0}^\infty$ such that*

1. *Consumers save a constant share of their income, $S_t = sY_t$, where s is an exogenous parameter. Note that consumers own all capital and labor.*¹
2. *Output-producing firms solve*

$$\max_{X_t, L_t} Y_t - R_t^X X_t - w_t L_t$$

taking the factor prices R_t^X and w_t as given. Y_t is given by $Y_t = X_t^\alpha L_t^{1-\alpha}$.

3. *Capital goods-producing firms solve*

$$\max_{I_t} p_t^X I_t^X(I_t) - I_t$$

where the production function is given by $I_t^X(I_t) = q_t I_t$

4. *All firms make zero profits*
5. *All markets clear, in particular,*

(a) *The final goods market*

$$Y_t = C_t + I_t$$

(b) *the labor market*

$$L_t^d = L_t^s$$

¹Note that in most other models this assumption on consumer behavior is replaced by a utility maximization problem.

(c) the capital market

$$X_t^d = X_t^s$$

6. Capital is accumulated according to

$$\dot{X}_t = I_t^X - \delta X_t, \quad \delta \in (0, 1),$$

labor supply grows exogenously at rate n so that

$$\dot{L}_t = nL_t$$

and the technology to produce capital goods improves at rate γ so that

$$\dot{q}_t = \gamma q_t.$$

Note that there is some redundancy in the description of the model and the equilibrium definition. This is on purpose: an equilibrium definition must contain all information that is relevant for the solution of the model.

2 Solution

2.1 Agents' problems

Consider first the output-producing firm with problem

$$\max_{X_t, L_t} X_t^\alpha L_t^{1-\alpha} - R_t^X X_t - w_t L_t.$$

First order conditions give

$$\alpha X_t^{\alpha-1} L_t^{1-\alpha} = R_t^X$$

and

$$(1 - \alpha) X_t^\alpha L_t^{-\alpha} = w_t$$

These relationships are the inverse factor demand functions for (physical) capital and labor.

Next, consider the capital goods-producing firm. Their problem is

$$\max_{I_t} p_t^X q_t I_t - I_t.$$

Such an objective looks pathological, but its solution is simple. The firms demand function for I_t , the amount of GDP spent on investment into physical capital, is given by

$$I_t^d = \begin{cases} \infty & \text{if } p_t^X q_t > 1 \\ [0, \infty) & \text{if } p_t^X q_t = 1 \\ 0 & \text{if } p_t^X q_t < 1 \end{cases}$$

Finally, consider the household's problem. This is really trivial, since we are told that $S_t = sY_t$. Then, by the households budget constraint $C_t + S_t = Y_t$ consumption is $C_t = (1 - s)Y_t$.

2.2 Market clearing

First note that output market clearing implies that

$$I_t^d = Y_t - C_t = Y_t - (1-s)Y_t = sY_t$$

This means that the household, who owns the capital stock wishes to supply sY_t units of GDP to firms that build physical capital. Since $0 < sY_t < \infty$ we know by (1) that we must be in the case in which $p_t^X q_t = 1$, or $p_t^X = q_t^{-1}$. Furthermore, from the production function we get that $I_t^X = q_t I_t = q_t s Y_t$. Finally, we note that $L_t^d = L_t^s$ and $X_t^d = X_t^s$ so that we can recover the factor prices of physical capital and labor for any given sequence of supplies L_t and X_t

$$\begin{aligned} R_t^X &= \alpha X_t^{\alpha-1} L_t^{1-\alpha} \\ w_t &= (1-\alpha) X_t^\alpha L_t^{-\alpha} \end{aligned}$$

2.3 Dynamics and interpretation

First, define the market value of capital as $K_t = p_t^X X_t = \frac{X_t}{q_t}$, that is price times quantity. I want to rewrite the aggregate production function as

$$\begin{aligned} Y_t &= X_t^\alpha L_t^{1-\alpha} \\ &= (q_t K_t)^\alpha L_t^{1-\alpha} \\ &= K_t^\alpha \left(q_t^{\frac{\alpha}{1-\alpha}} L_t \right)^{1-\alpha}. \end{aligned}$$

This is a production function in which technological progress is expressed in a labor-augmenting form.

Next, we plug $I_t^X = q_t s Y_t = q_t s X_t^\alpha L_t^{1-\alpha}$ into the capital accumulation equation to get

$$\dot{X}_t = q_t s X_t^\alpha L_t^{1-\alpha} - \delta X_t.$$

This equation together with

$$\dot{L}_t = n L_t$$

and

$$\dot{q}_t = \gamma q_t$$

governs the complete dynamics of the model for any given initial values q_0 , L_0 , and X_0 . However, due to the magic of the Solow model we don't need to deal with a system of three differential equations simultaneously. It turns out that it is sufficient to look at an artificial variable:

$$k_t := \frac{K_t}{q_t^{\frac{\alpha}{1-\alpha}} L_t}$$

From here, take logs to get

$$\ln(k_t) = \ln(K_t) - \frac{\alpha}{1-\alpha} \ln(q_t) - \ln(L_t).$$

Now take derivatives w.r.t. time so that

$$\frac{\dot{k}_t}{k_t} = \frac{\dot{K}_t}{K_t} - \frac{\alpha}{1-\alpha} \frac{\dot{q}_t}{q_t} - \frac{\dot{L}_t}{L_t}$$

Recognizing that $K_t = \frac{X_t}{q_t}$ we get in a similar fashion

$$\frac{\dot{K}_t}{K_t} = \frac{\dot{X}_t}{X_t} - \frac{\dot{q}_t}{q_t}.$$

Next, substitute what we know about these growth rates to get

$$\begin{aligned} \frac{\dot{k}_t}{k_t} &= \frac{\dot{X}_t}{X_t} - \frac{\dot{q}_t}{q_t} - \frac{\alpha}{1-\alpha} \frac{\dot{q}_t}{q_t} - \frac{\dot{L}_t}{L_t} \\ &= \frac{\dot{X}_t}{X_t} - \left(\frac{1}{1-\alpha} \right) \frac{\dot{q}_t}{q_t} + \frac{\dot{L}_t}{L_t} \\ &= q_t s X_t^{\alpha-1} L_t^{1-\alpha} - \delta - \left(\frac{1}{1-\alpha} \right) \gamma + n \\ &= s \left(\frac{q_t^{\frac{1}{1-\alpha}} X_t}{L_t} \right)^{\alpha-1} - \delta - \left(\frac{1}{1-\alpha} \right) \gamma + n \\ &= s \left(\frac{q_t^{\frac{1}{1-\alpha}} q_t K_t}{L_t} \right)^{\alpha-1} - \left(\delta + \frac{1}{1-\alpha} \gamma + n \right) \\ &= s \left(\frac{K_t}{q_t^{\frac{\alpha}{1-\alpha}} L_t} \right)^{\alpha-1} - \left(\delta + \frac{1}{1-\alpha} \gamma + n \right) \end{aligned}$$

and hence we get

$$\dot{k}_t = s k_t^\alpha - \left(\delta + \frac{1}{1-\alpha} \gamma + n \right) k_t$$

All dynamics of the model can be recovered from this differential equation. If we let $d = \delta + \frac{1}{1-\alpha} \gamma$ we obtain

$$\dot{k}_t = s k_t^\alpha - (d + n) k_t$$

for which we have shown that there is a unique nontrivial steady state and monotone convergence to this steady state from any starting value $k_0 > 0$. Therefore, the dynamics of the transformed model remain the same.

What is the interpretation of d ? The formula suggests a decomposition of depreciation into obsolescence from technological progress and physical wear and tear:

$$d = \underbrace{\delta}_{\text{wear and tear}} + \underbrace{\frac{1}{1-\alpha} \gamma}_{\text{obsolescence}}$$

3 Digression: the Bernoulli differential equation

The equation

$$\dot{k}_t = sk_t^\alpha - (d+n)k_t \quad (2)$$

is a nonlinear differential equation. It is a special case of a Bernoulli differential equation which can be made linear through an appropriate transformation. Consider the transformation $x_t = k_t^{1-\alpha}$. Then

$$\begin{aligned} \dot{x}_t &= (1-\alpha)k_t^{-\alpha}\dot{k}_t \\ \frac{\dot{x}_t}{k_t^\alpha(1-\alpha)} &= \dot{k}_t \end{aligned}$$

Substitution gives

$$\begin{aligned} \frac{\dot{x}_t}{k_t^\alpha(1-\alpha)} &= sk_t^\alpha - (d+n)k_t \\ \dot{x}_t &= (1-\alpha)s - (1-\alpha)(d+n)k_t^{1-\alpha} \\ \dot{x}_t &= (1-\alpha)s - (1-\alpha)(d+n)x_t \end{aligned}$$

which is linear.

Now the associated homogenous differential equation is

$$\dot{x}_t = -(1-\alpha)(d+n)x_t$$

with general solution

$$x_t^h = Ce^{-(1-\alpha)(d+n)t}$$

for some C .

One particular solution is the stationary one where $\dot{k}_t = \dot{x}_t = 0$. This condition gives

$$\begin{aligned} 0 &= (1-\alpha)s - (1-\alpha)(d+n)x^* \\ x^* &= \frac{s}{d+n} \end{aligned}$$

The solution to the inhomogenous ODE is then

$$x_t = x_t^h + x^* = Ce^{-(1-\alpha)(d+n)t} + \frac{s}{d+n}$$

Substituting back gives

$$k_t = \left(Ce^{-(1-\alpha)(d+n)t} + \frac{s}{d+n} \right)^{\frac{1}{1-\alpha}}$$

For an initial value k_0 we get

$$k_0^{1-\alpha} - \frac{s}{d+n} = C$$

so that the unique solution to the inhomogenous ODE in k given by (2) and a starting value k_0 is

$$k_t = \left(\left(k_0^{1-\alpha} - \frac{s}{d+n} \right) e^{-(1-\alpha)(d+n)t} + \frac{s}{d+n} \right)^{\frac{1}{1-\alpha}}$$